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PREDICTING MACROALGAL SHIFTS: CLIMATE AND ENVIRONMENTAL DRIVERS IN THE CANARY ISLANDS (NE ATLANTIC OCEAN) OVER 28 YEARS

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Abstract:	Macroalgal communities are sensitive to environmental changes, making them key indicators for assessing climate change impacts on marine biodiversity. This study examines the relationship between environmental variables and the relative abundance of macroalgal orders Bryopsidales, Ceramiales, Dictyotales, and Fucales across the Canary Islands over 28 years. Significant multi-variable correlations were found, reflecting temporal environmental changes. While some models explained substantial variance, high environmental heterogeneity across the archipelago reduced model precision. Grouping islands into eastern and western regions and applying Generalized Additive Models (GAMs) improved predictive accuracy, emphasizing the value of region-specific modeling approaches. Principal Component Analysis (PCA) revealed an environmental gradient influencing macroalgal distributions, with nutrient-rich eastern islands showing stronger correlations for certain macroalgae. Environmental factors such as sea surface temperature, salinity, and nutrient levels were key drivers of distribution, varying by region and macroalgal order. For example, Bryopsidales and Ceramiales showed stable responses in nutrient-rich eastern waters, while temperate species in western islands were more sensitive to warming, highlighting vulnerability to climate change. The findings highlight the need for predictive models integrating environmental variables and ecological interactions, particularly in heterogeneous regions like the Canary Islands. Incorporating fine-scale biogeochemical data into future models will enhance our understanding of species shifts under climate change, identifying areas at risk of biodiversity loss and tropicalization. Long-term satellite data and ecological records are essential for devising effective conservation strategies to safeguard marine biodiversity in response to climate-driven changes.
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Author Comments:	Dear Editor-in-Chief, We are pleased to submit our manuscript titled "Predicting Macroalgal Shifts: Climate and Environmental Drivers in the Canary Islands (NE Atlantic Ocean) Over 28 Years" for consideration in Regional Environmental Change. This study addresses pressing questions about the impact of climate and environmental changes on marine biodiversity, using macroalgal communities in the Canary Islands as sentinel systems. Macroalgal communities serve as critical indicators of environmental shifts, and our 28-year dataset offers unique insights into how sea surface temperature, salinity, and nutrient levels drive changes in macroalgal abundance and distribution. By employing region-specific Generalized Additive Models (GAMs) and Principal Component Analysis (PCA), we identify key environmental gradients and their influence on macroalgal orders across this heterogeneous archipelago. Our findings underscore the importance of incorporating fine-scale biogeochemical data and predictive modeling to forecast climate change impacts and guide conservation strategies. Our research aligns with the journal's focus on advancing regional understandings of environmental change and its ecological consequences. Specifically, our study contributes to: Understanding the drivers of marine biodiversity shifts under climate change; (i) Highlighting the role of environmental heterogeneity in influencing model precision and predictive accuracy; (ii) Offering actionable insights for biodiversity conservation in marine ecosystems vulnerable to warming and tropicalization. We believe this manuscript will interest your readers, including scientists and policymakers focused on climate-driven ecological changes and conservation. The manuscript has not been published elsewhere and is not under consideration by any other journal. We appreciate your time and consideration and look forward to the opportunity to contribute to Regional Environmental Change. Please do not hesitate to contact me via email at [your email] or by phone at [your phone number] should you require additional information. Thank you for considering our manuscript. We look forward to your feedback. Sincerely, The authors

PREDICTING MACROALGAL SHIFTS: CLIMATE AND ENVIRONMENTAL DRIVERS IN THE CANARY ISLANDS (NE ATLANTIC OCEAN) OVER 28 YEARS

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ABSTRACT

Macroalgal communities are sensitive to environmental changes, making them key indicators for assessing climate change impacts on marine biodiversity. This study examines the relationship between environmental variables and the relative abundance of macroalgal orders Bryopsidales, Ceramiales, Dictyotales, and Fucales across the Canary Islands over 28 years. Significant multi-variable correlations were found, reflecting temporal environmental changes. While some models explained substantial variance, high environmental heterogeneity across the archipelago reduced model precision. Grouping islands into eastern and western regions and applying Generalized Additive Models (GAMs) improved predictive accuracy, emphasizing the value of region-specific modeling approaches. Principal Component Analysis (PCA) revealed an environmental gradient influencing macroalgal distributions, with nutrient-rich eastern islands showing stronger correlations for certain macroalgae. Environmental factors such as sea surface temperature, salinity, and nutrient levels were key drivers of distribution, varying by region and macroalgal order. For example, Bryopsidales and Ceramiales showed stable responses in nutrient-rich eastern waters, while temperate species in western islands were more sensitive to warming, highlighting vulnerability to climate change. The findings highlight the need for predictive models integrating environmental variables and ecological interactions, particularly in heterogeneous regions like the

Canary Islands. Incorporating fine-scale biogeochemical data into future models will enhance our understanding of species shifts under climate change, identifying areas at risk of biodiversity loss and tropicalization. Long-term satellite data and ecological records are essential for devising effective conservation strategies to safeguard marine biodiversity in response to climate-driven changes.

Keywords: Macroalgae distribution, environmental variability, climate change, predictive modeling, oceanic islands.

INTRODUCTION

Climate change has introduced simultaneous shifts in temperature, ocean circulation, and stratification, producing significant and potentially widespread negative effects on marine ecosystems ([Waycott et al., 2009](#); [Brothers et al., 2018](#); [Turschwell et al., 2021](#)). Rising ocean temperatures contribute to sea level rise ([Jevrejeva et al., 2008](#); [Alcántara et al., 2022](#)), reduced sea-ice extent ([Hunt et al., 2022](#)), and changes in ocean circulation ([Noisette et al., 2022](#)), all of which drive accelerated degradation of marine habitats. Coastal systems, in particular, are experiencing severe loss, with global declines including 50% of salt flats, 35% of mangroves, 30% of coral reefs, and 29% of seagrasses ([Jackson, 2010](#)). These environmental changes force marine species to adapt, migrate, or face local extinction if conditions exceed their tolerance ([Parmesan, 2006](#); [Somero, 2010, 2012](#)). In oceanic ecosystems, shifts in species distributions due to these changes are altering community structures and biodiversity ([Baselga, 2013](#)), with notable impacts on foundational species like ecosystem-forming seaweeds ([Martínez et al., 2017](#)), which are critical to the stability of coastal ecosystems.

Predicting how habitat-forming macroalgae will respond to climate change is essential due to their role in providing food, shelter, and stabilizing ecosystem dynamics (Halpern et al., 2007; Thomsen et al., 2010). Shifts in these macroalgae populations, such as potential acclimatization or declines, have already been documented in several ecologically significant species (Takolander et al., 2017; Riera et al., 2015; Franco et al., 2016). Long-term datasets, such as herbarium records and remote sensing data, are valuable in studying these changes over decades, helping distinguish climate-driven trends from natural variability (Ducklow et al., 2009; Beck, 2017; McClain, 2009). Integrating these biological data with environmental metrics enables the application of remote sensing techniques to monitor and predict shifts in habitat-forming species distributions under changing climate conditions (Martínez et al. 2018; Kanavaugh et al. 2021).

Generalized Additive Models (GAMs) are particularly useful for analyzing long-term datasets, as they allow for the exploration of non-linear relationships between environmental variables and species distributions (Hastie et al. 2017). GAMs can integrate multiple predictor variables to model species responses to environmental factors, providing insights into the conditions that influence their distribution and abundance (Yee & Mitchell, 1991; Stoner et al., 2011). These models have proven effective in studying complex ecosystems, but their accuracy often depends on the availability of extensive, region-specific data, which can be challenging to acquire (Grim, 1994; Drexler & Ainsworth, 2013). The Canary Islands, with their distinct oceanographic and ecological characteristics, offer an ideal location for applying GAMs to study marine and terrestrial ecosystems. The pronounced environmental gradient across the archipelago, shaped by the variability in Canary currents (Barton et al. 1998; Noisette et al., 2022), creates a natural laboratory for exploring non-linear species-environment relationships. From the nutrient-rich waters of the eastern islands like Lanzarote and Fuerteventura to the warmer, more stable

ecosystems of western islands such as La Palma and El Hierro, the region provides diverse and contrasting conditions that can be modeled effectively using GAMs. Additionally, the islands' geographic, climatic, and geological gradients ([Hanz et al., 2022](#)) enable the study of species with restricted geographic distributions. Their wealth of unique habitats and long-term environmental datasets enhances our understanding of how local and regional ecological factors influence biodiversity, despite challenges posed by anthropogenic pressures in this archipelago ([Riera et al. 2014](#)), considering the spatial scale in the studied regions ([MacArthur et al. 1992](#))

In this study, we use herbarium records of four macroalgae orders (Bryopsidales, Ceramiales, Dictyotales, and Fucales) from the Canary Islands to explore environmental interactions affecting macroalgal populations over the past three decades (1993–2020). As ecosystem engineers, these macroalgae support biodiversity and stabilize coastal habitats in the Canary Islands (e.g., [Valdazo et al. 2017](#); [Casado-Abellán & Riera, 2024](#); [González-Aragón et al. 2024](#)). They enhance ecosystem resilience by providing habitat and contributing to nutrient cycling, which sustains marine species and supports local fisheries ([Mineur et al. 2015](#)). By analyzing species abundance against temporal records of sea surface temperature, salinity, and height for the entire archipelago, individual islands, and regional groupings (Western and Eastern islands), we aim to model the dependence of these macroalgae orders on climate-sensitive environmental factors using GAMs. This approach will provide insights into keystone macroalgae's vulnerability and adaptive responses in the Canarian coastal ecosystems.

MATERIAL AND METHODS

Study area

The study area defined for this study is the Canary Archipelago (NE Atlantic Ocean). The data used in this study were collected separately for each island, allowing for distinct information on each island to be analyzed independently. In addition, considering the oceanographic characteristics and the environmental gradient that the archipelago presents ([Barton et al., 1998](#); [Riera & Delgado, 2019](#)), the area was classified into two large groups of islands. The western islands, conformed by El Hierro, La Palma, La Gomera and Tenerife, with tropical affinities in their marine flora ([Tuya et al. 2004](#)) (Fig. 1). Eastern islands conformed by Gran Canaria, Lanzarote and Fuerteventura characterized by a cold-temperate flora ([Tuya et al. 2004](#)).

Macroalgae records

A total of 11,304 records of four macroalgae orders were used. Namely, 2,157 records of Bryopsidales; 4,081 records of Ceramiales; 1,784 records of Dictyotales; and 3,282 records of Fucales. These specimens were collected in 5,437 locations throughout the seven islands of the Canary archipelago in a 28-year time interval (from 1992 to 2020) (Fig 2). The samples used were obtained from the records of the Canary Island Biodiversity Data Bank ([Government of the Canary Islands, 2022](#)) and some records deposited in the Herbarium BCM database from the Marine Science Faculty ([University of Las Palmas de Gran Canaria, 2021](#))

Environmental data

Environmental variables from the Copernicus Marine Service ([Dréville et al. 2021](#)) were utilized in this study. The variables chosen were the Sea Surface Temperature (°C) (SST), Sea Surface Salinity (PSU), Significant Wave Height (m)(SWH), Sea Surface Radiation (J m^{-2})(SSR), nitrate

($\mu\text{M/L}$)(NO_3), and phosphate ($\mu\text{M/L}$)(PO_4), due to the relevant effects produced on macroalgae communities in the ocean (Sunny, 2017). These data were collected in netCDF.nc format with a spatial resolution of $1/12^\circ$ per pixel (ca. $8\text{km}^2/\text{pixel}$), a monthly average of data as temporal resolution (Drévellion et al. 2021) and a temporal range of 28 years according to the macroalgae records.

Two sets of environmental variables were prepared: one specific to each island, and another distinguishing between western (El Hierro, La Palma, La Gomera, and Tenerife) and eastern islands (Gran Canaria, Lanzarote, and Fuerteventura) to account for the environmental gradient across the region (Tuya et al. 2004) (Fig. 3).

Data processing

To minimize spatial and temporal biases due to heterogeneity in macroalgae records, specimens were filtered by species, sampling location, and year. This process retained only one record per species at each sampling point per year (Fig. 2), preventing multiple samples of the same macroalgae at the same location and time from artificially inflating the total record.

Once the macroalgae data was filtered, a count matrix was created by each order, island, and year. From this matrix, only orders with more than 1,500 records across the 28-year study period (1992-2020) were included in the analysis (Table 1). These count values of selected orders per year were treated as relative abundances for each island, enabling the assessment of macroalgal abundance trends throughout the archipelago over time (Fig. 2). Additionally, the total macroalgae records for each year were represented as frequencies to evaluate the distribution patterns of the response variable for each order (Bryopsidales, Ceramiales, Dictyotales, and Fucales) (Fig. 3).

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4 With the data visualization program NASA GISS: Panoply netCDF, HDF, and GRIB Data
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6 Viewer the environmental information was treated to recapitulate area averages and convert
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8 satellite image records into 28 year time series. A filtering treatment was applied to these time
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10 series according to the sampling frequency using the function `decompose {stats::decompose()}`
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12 from the package STATS V3.6.2 in RStudio ([R Core Team, 2022](#)). This filtering had the purpose
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14 of eliminating the residual components of the time series, obtaining the trend and stationarity of
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16 each variable over time more clearly. Once the time series were treated, the information was
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18 assigned to the macroalgae counting records by statistical model selection. In this way, the
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20 macroalgae counts will have added the environmental measurements for spatial region and year.
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27 ***Statistical analysis***

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30 For the statistical analysis, we first establish relationships between environmental
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32 variables alongside the time series of macroalgal relative abundance to visually assess potential
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34 relationships by Principal Component Analysis (PCA) (Fig. 4). Given the temporal variability and
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36 environmental heterogeneity across the archipelago, linear correlations between the dependent
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38 variable (relative abundance) and the independent variables (environmental drivers) were not
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40 apparent. Therefore, a Generalized Additive Model (GAM) was selected for the analysis to account
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42 for non-linear relationships and temporal trends ([Wood et al. 2017](#)).
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49 GAM models were applied to examine the environmental effects on relative abundance
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51 across the entire archipelago (Table 2). Second, separate GAM models were developed for each
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53 island (Table 3). Finally, GAM models were created for the two island groups: Western and
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55 Eastern islands (Table 4). The GAM analysis was performed in R Studio version 2023.9.1.494 ([R
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[Core Team 2022](#)), using cubic regression splines (“cr”) with a smoothing parameter (k) of 30 for

each environmental variable. The Tweedie family ($p = 1.5$) with a logarithmic link function was used, as it is well-suited for data comprising zeros and positive values, particularly when dealing with highly dispersed distributions. The model was fitted using the REML method for stable smoothing parameter estimation, and adaptive penalization allowed the removal of non-informative predictors. In addition to the aforementioned environmental variables, *Year* was included as a categorical factor to capture inter-annual variability.

The analysis was conducted in R Studio using the mgcv package (Wood et al. 2017, Pedersen et al. 2019). The different models were chosen by ANOVA comparison between several model designs, selecting the greatest explanatory capacity and the least defined complexity.

RESULTS

Analysis of the correlations between environmental variables and macroalgal relative abundance records across the Canary Islands landscape revealed significant multi-variable relationships for each macroalgal order, corresponding to temporal variability in environmental factors. Temporal changes in environmental components accounted for 54.9% to 66.1% of the variation in macroalgal relative abundance over the 28-year time series. Despite these significant relationships, model fits did not show consistently strong alignment, likely due to high environmental variability in the Canary Islands, which affects model precision even with significant correlations between predictor and response variables (Table 2).

For Bryopsidales across the archipelago, environmental factors explained 66.8% of the variance with four key variables, though the correlation adjustment remained low at 35%.

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4 Similarly, Ceramiales variance was explained at 61.5% with five variables but yielded a moderate
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6 correlation of 43.9%. Dictyotales exhibited 64.5% data variance explained by five significant
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8 variables and a correlation adjustment of 30.4%, while Fucales records showed 54.9% variance
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10 explained by two environmental variables with a 28.9% correlation.
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15 PCA indicated regional differentiation, with a gradient from positive PC1 values in the
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17 western islands (e.g., La Palma and El Hierro) to negative PC1 values in the eastern islands (Fig.
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19 4). This regional gradient suggests that temporal series should be analyzed on an island-by-island
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21 basis to account for region-specific environmental variables affecting macroalgal distributions.
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23 The model selection process led to the establishment of Generalized Additive Models (GAMs) for
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25 each macroalgal order per region. Despite using the same variables for all models, regional
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27 variability caused significant differences in fit and explained variance, likely due to both sample
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29 heterogeneity and environmental variation across islands. The observed differences in R^2 values
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31 and explanatory deviation highlight the influence of regional environmental heterogeneity on
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33 model accuracy (Table 3).
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40 When records were grouped into two broad regions—western islands (El Hierro, La Palma,
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42 La Gomera, and Tenerife) and eastern islands (Lanzarote, Fuerteventura, and Gran Canaria)—
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44 environmental variables showed greater model fit and explanatory power than models-based on
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46 individual islands or the entire archipelago (Table 4). This grouping revealed that specific
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48 environmental variables impacted macroalgal orders differently in each region.
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54 For Bryopsidales in the western islands, three environmental variables, i.e., Sea Surface
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56 Temperature (SST), Sea Surface Radiation (SSR), and Salinity explained 85% of the variance with
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58 a correlation of 0.763. In the eastern islands, significant variables included SSR, Salinity, and
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4 Significant Wave Height (SWH), achieving a stronger fit with a 92.5% variance explained and a
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6 correlation of 0.852. For Ceramiales, in the western region, four environmental factors (SST,
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8 SWH, Nitrate (NO_3), and Phosphate (PO_4)) explained 84.9% of the variance with a 0.77
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10 correlation. In the eastern islands, only three variables were necessary (excluding PO_4), yielding
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12 90.5% variance explained and a 0.782 correlation. For Dictyotales, variance explained reached
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14 77.5% in the west with three variables (SST, Salinity, PO_4) but showed a lower correlation of
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16 0.511, likely reflecting fewer samples. In the eastern region, five variables (SSR, SWH, Salinity,
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18 NO_3 , and PO_4) improved the variance explained to 82.7% with a 0.733 correlation. Finally, for
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20 Fucales, two environmental variables (SST and SSR) in the western region explained 81.6% of the
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22 variance with a correlation of 0.79. In the east, the addition of SWH and PO_4 resulted in a slightly
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24 lower variance explained (73.5%) and a correlation of 0.653.
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32 Using island-specific models and grouping islands into two main regions—western (El
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34 Hierro, La Palma, La Gomera, and Tenerife) and eastern islands (Lanzarote, Fuerteventura, and
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36 Gran Canaria)—improved model fit and accuracy, as environmental factors influenced macroalgal
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38 orders more predictably within each region. In the western islands, Bryopsidales variance was
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40 explained at 85% by SST, Sea Surface Radiation (SSR), and Salinity, with a correlation of 0.763.
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42 In the eastern islands, the inclusion of SSR, Salinity, and Significant Wave Height (SWH) yielded
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44 a stronger fit, with 92.5% variance explained and a correlation of 0.852. For Ceramiales, four
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46 variables (SST, SWH, NO_3 , and PO_4) explained 84.9% of the variance in the western region, while
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48 in the eastern region, three factors excluding PO_4 improved the variance explained to 90.5%, with
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50 a correlation of 0.782.
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60 DISCUSSION

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7 The main results of this study illustrate a clear decline in macroalgal populations across the Canary
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9 Islands, which is primarily driven by human-driven activities and environmental factors showing
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11 considerable spatial and temporal variability. Significant differences exist between the western and
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13 eastern islands, including SST, nutrient concentrations, and salinity levels, each of which
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15 influences macroalgal distributions differently (Tuya et al. 2006; Barton et al. 1998). The
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17 environmental characteristics of the Canary archipelago—observed through satellite data—
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19 emphasize that environmental components distinguish regions and contribute to contrasting
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21 macroalgal dynamics at a regional level across time.
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26 Analyses of environmental variables against long-term macroalgal relative abundance
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28 records indicate significant multi-variable relationships for each macroalgal order across each
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30 island, reflecting temporal changes in environmental conditions. Temporal variability accounted
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32 for 54.9% to 66.1% of the variance in macroalgal relative abundance over a 28-year time series.
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34 However, high environmental variability across the archipelago limits model accuracy, reflected
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36 in moderate correlation adjustments despite significant predictor-response relationships. For
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38 instance, the Bryopsidales model explained 66.8% of the variance with four significant variables
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40 but had only a 35% correlation, revealing challenges due to regional heterogeneity.
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46 The present results align with previous research showing that environmental stressors
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48 disproportionately impact macroalgal populations based on local tolerance thresholds for SST,
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50 salinity, and nutrient availability (Zou and Gao, 2013; Wernberg et al. 2012; Staehr and Wernberg,
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52 2009; Pekker et al. 1996). Former studies revealed that environmental heterogeneity in the Canary
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54 Islands influences the connectivity and dispersal of temperate-water macroalgae, such as Fucales,
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56 which face declines due to increased SST and SSH in warmer waters of the western islands (Duarte
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4 [et al. 2013; Riera et al. 2015](#)). In contrast, lower temperatures and higher nutrient input in the
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6 eastern islands provide a temporary buffer against warming impacts, favoring specific taxa
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9 ([Afonso et al. 2013; Ribeiro et al. 2019](#)).
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13 GAMs emphasized the significance of regional environmental variability in shaping
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15 macroalgal distributions, with island grouping models outperforming those for individual islands
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17 or the entire archipelago by capturing localized heterogeneity. This aligns with PCA, which
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19 revealed an environmental gradient: positive PC1 values in the western islands and negative values
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21 in the eastern islands, driven by distinct regional factors. Bryopsidales and Ceramiales, for
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23 instance, showed stronger associations with nutrient-rich, stable eastern waters. While these
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25 models provide insights into environmental influences on macroalgal abundance, they do not fully
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27 predict species metacommunity dynamics due to the complex interplay of species traits and
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29 ecological interactions. Reliable predictions require integrating biological characteristics with
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31 environmental factors through frameworks like species distribution modeling. Spatial
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33 heterogeneity in the Canary Islands further complicates predictions, with unique temporal
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35 fluctuations at each site limiting model performance ([Riera et al. 2015](#)). Broader data collection
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37 and localized assessments are essential to complement large-scale models, addressing fine-scale
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39 variability and accurately reflecting species dynamics in heterogeneous landscapes ([MacArthur, et](#)
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41 [al. 1992; Zou & Gao, 2013](#)).
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51 As climate change progresses, the variability in SST, SSH, and Salinity could exacerbate
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53 regional declines in macroalgal diversity, particularly in western islands, where populations
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55 already display sensitivity to warming ([Valdazo et al. 2024](#)). These shifts may allow species with
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57 tropical affinities to expand, while temperate species decline ([Perry et al. 2005; Martínez et al.](#)
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59 [2017; Franco et al. 2018](#)). If warming continues as projected by the [IPCC \(2020\)](#), this shift could
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4 trigger cascading effects on ecosystem function and biodiversity ([Smale and Wernberg, 2013](#);
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6 [Bracken et al. 2007](#)), potentially leading to local extinctions due to limited adaptation capacity
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8 linked to genetic diversity loss ([Pearson et al. 2009](#)). As observed in other studies, a decrease in
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10 habitat-specific species could favor tropical species suited to warmer conditions, reshaping
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12 community composition ([Vergés et al. 2016](#); [Valdazo et al. 2017](#); [Ribeiro et al. 2019](#)).
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16 The present study reinforces the importance of developing statistical models that
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18 incorporate both physical and biogeochemical interactions to better predict long-term impacts on
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20 marine ecosystems. Enhanced regional models integrating SST, salinity, nutrient levels, and other
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22 variables could provide clearer insights into climate change effects and support targeted
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24 conservation strategies ([Martínez et al. 2012](#); [Bennett et al. 2018](#)). These models should also
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26 identify climate refugia—areas that are less affected by climate change and can serve as safe
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28 havens for species under stress. Preserving these refugia is vital for sustaining native macroalgal
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30 species in the Canary Islands and ensuring their role in maintaining marine biodiversity ([Riera &](#)
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32 [Delgado, 2019](#)). By prioritizing the protection of regions with stable environmental conditions and
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34 high resilience to climate shifts, conservation efforts can help mitigate the adverse effects of
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36 climate change. This approach aligns with calls for adaptive management strategies to safeguard
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38 marine biodiversity under the growing climate crisis ([Wernberg et al. 2011](#); [Riera et al. 2015](#);
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40 [Vergés et al. 2014, 2016](#)). Identifying and preserving such critical areas will enhance ecosystem
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42 stability and offer a foundation for the recovery of vulnerable species.
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53 CONCLUSIONS

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55 Integrating satellite data with long-term biological records has provided valuable insights into how
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57 the unique environmental conditions of the Canary archipelago shape macroalgal tolerance and
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4 distribution. This research underscores the importance of incorporating diverse environmental
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6 variables into population studies, particularly in complex marine ecosystems like the Canaries.
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8 Scaling up these findings to other regions requires the development of spatially consistent and
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10 temporally continuous sampling frameworks, enabling more precise and globally relevant
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12 predictions for conservation.
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17 Future research should prioritize identifying areas most vulnerable to climate impacts or
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19 those that could serve as refugia for native species. Incorporating detailed biogeochemical
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21 variables, high-resolution datasets, and refined ecological models will be crucial for accurately
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23 predicting species dynamics and pinpointing regions susceptible to biodiversity shifts, including
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25 the establishment of invasive tropical species. Conservation strategies must focus on mitigating
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27 the loss of macroalgal populations by safeguarding critical habitats essential for their persistence.
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33 This study highlights the broader applicability of its methods and findings, providing a
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35 foundation for predictive modeling in other biodiverse and environmentally heterogeneous
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37 regions. Proactive measures are essential to preserve biodiversity in the face of ecosystem shifts
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39 driven by climate change. Escalating these predictive models can inform adaptive management
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41 plans to diverse marine landscapes, ensuring the protection of both macroalgal communities and
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43 the broader ecosystems they support. By refining these approaches, researchers and policymakers
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45 can develop robust conservation frameworks to address the global challenges posed by climate-
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47 driven changes in marine biodiversity.
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52 53 54 55 56 57 58 59 60 **ACKNOWLEDGEMENTS** 61 62 63 64 65

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TABLES AND FIGURES

Table 1: Number of algae samples from the four macroalgae orders (Bryopsidales, Ceramiales, Dictyotales and Fucales) selected for the entire Canary Islands region (Totals) and for each island of the archipelago [El Hierro (EH), La Palma (LP), La Gomera (LG), Tenerife (TF), Gran Canaria (GC), Lanzarote (LZ) and Fuerteventura (FV)].

Order	EH	LP	LG	TF	GC	LZ	FV	Total
Bryopsidales	45	50	140	255	880	503	284	2157
Ceramiales	117	364	17	680	885	564	1454	4081
Dictyotales	38	122	11	431	278	763	141	1784
Fucales	42	135	39	430	2190	291	155	3282
Total	242	671	207	1796	4233	2121	2034	11304

Table 2. Generalized additive model (GAM) results for four macroalgae orders (Bryopsidales, Ceramiales, Dictyotales, and Fucales) in the Canary Islands, assessing their response to environmental factors such as Salinity, Sea Surface Temperature (SST), Significant Wave Height (SWH), Surface Solar Radiation (SSR), Nitrate (NO₃), and Phosphate (PO₄).

Order	Significant variables	R ² (Adj)	Deviance explained
Bryopsidales	SST, SWH, Salinity	0.35	66.8%
Ceramiales	SST, SSR, SWH, NO ₃ , PO ₄	0.439	61.5%
Dictyotales	SST, SWH, Salinity, NO ₃ , PO ₄	0.304	64.5%
Fucales	SSR, Salinity	0.289	54.9%

Table 3. Generalized additive model (GAM) results for four macroalgae orders (Bryopsidales, Ceramiales, Dictyotales, and Fucales) in the seven islands of Canary Islands, assessing their response to environmental factors such as Salinity, Sea Surface Temperature (SST), Significant Wave Height (SWH), Surface Solar Radiation (SSR), Nitrate (NO₃), and Phosphate (PO₄).

Island	Order	Significant variables	R ² (Adj)	Deviance explained
El Hierro	Bryopsidales	-	0.037	16.1%
	Ceramiales	SWH, NO ₃	0.76	75.3%
	Dictyotales	SSR	0.601	70.2%
	Fucales	SSR, PO ₄	0.296	59.1%
La Palma	Bryopsidales	Salinity, NO ₃	0.654	76.7%
	Ceramiales	SST	0.25	47.3%
	Dictyotales	SSR, SWH	0.491	73.5%
	Fucales	SST, NO ₃	0.75	78.5%
La Gomera	Bryopsidales	Salinity	0.878	81.2%
	Ceramiales	NO ₃	0.13	41%
	Dictyotales	SSR	0.522	79.2%
	Fucales	SST, SWH, NO ₃	0.796	86.1%
Tenerife	Bryopsidales	SST, SSR	0.678	60.2%

Gran Canaria	Ceramial	SST, NO ₃	0.213	38.9%
	Dictyotal	SSR, NO ₃ , PO ₄	0.664	53.9%
	Fucal	Salinity, PO ₄ ,	0.182	35.6%
	Bryopsidales	SSR, SWH	0.6	65.3%
	Ceramiales	SST, SWH, Salinity, NO ₃	0.531	48.9%
Lanzarote	Dictyotales	SST, SWH, Salinity, PO ₄	0.748	73.9%
	Fucales	SST, PO ₄	0.6	70.7%
	Bryopsidales	SST, SWH, NO ₃	0.431	62.1%
	Ceramiales	Salinity, PO ₄	0.815	68.4%
	Dictyotales	SST, NO ₃	0.359	51.8%
Fuerteventura	Fucales	SST, SWH	0.217	55%
	Bryopsidales	SSR, SWH, Salinity	0.667	73.8%
	Ceramiales	NO ₃ , PO ₄	0.69	62.8%
	Dictyotales	SST	0.466	52.6%
	Fucales	SST, Salinity	0.6	68.7%

Table 4. Generalized additive model (GAM) results for four macroalgae orders (Bryopsidales, Ceramiales, Dictyotales, and Fucales) in the western islands of Canary Islands (El Hierro, Las Palmas, La Gomera, and Tenerife) and the eastern islands of Canary Islands (Gran Canaria, Lanzarote, and Fuerteventura), assessing their response to environmental factors such as Salinity, Sea Surface Temperature (SST), Significant Wave Height (SWH), Surface Solar Radiation (SSR), Nitrate (NO₃), and Phosphate (PO₄).

Island group	Order	Significant variables	R2 (Adj)	Deviance explained
Western	Bryopsidales	SST, SSR, Salinity	0.763	85%
	Ceramiales	SST, SWH, NO ₃ , PO ₄	0.77	84.9%
	Dictyotales	SST, Salinity, PO ₄	0.511	77.5%
	Fucales	SST, SSR	0.79	81.6%
Eastern	Bryopsidales	SWH, SSR, Salinity	0.852	92.5%
	Ceramiales	SST, SSR, NO ₃	0.782	90.5%
	Dictyotales	SSR, SWH, Salinity, NO ₃ , PO ₄	0.733	82.7%
	Fucales	SST, SSR, SWH, PO ₄	0.653	73.5%

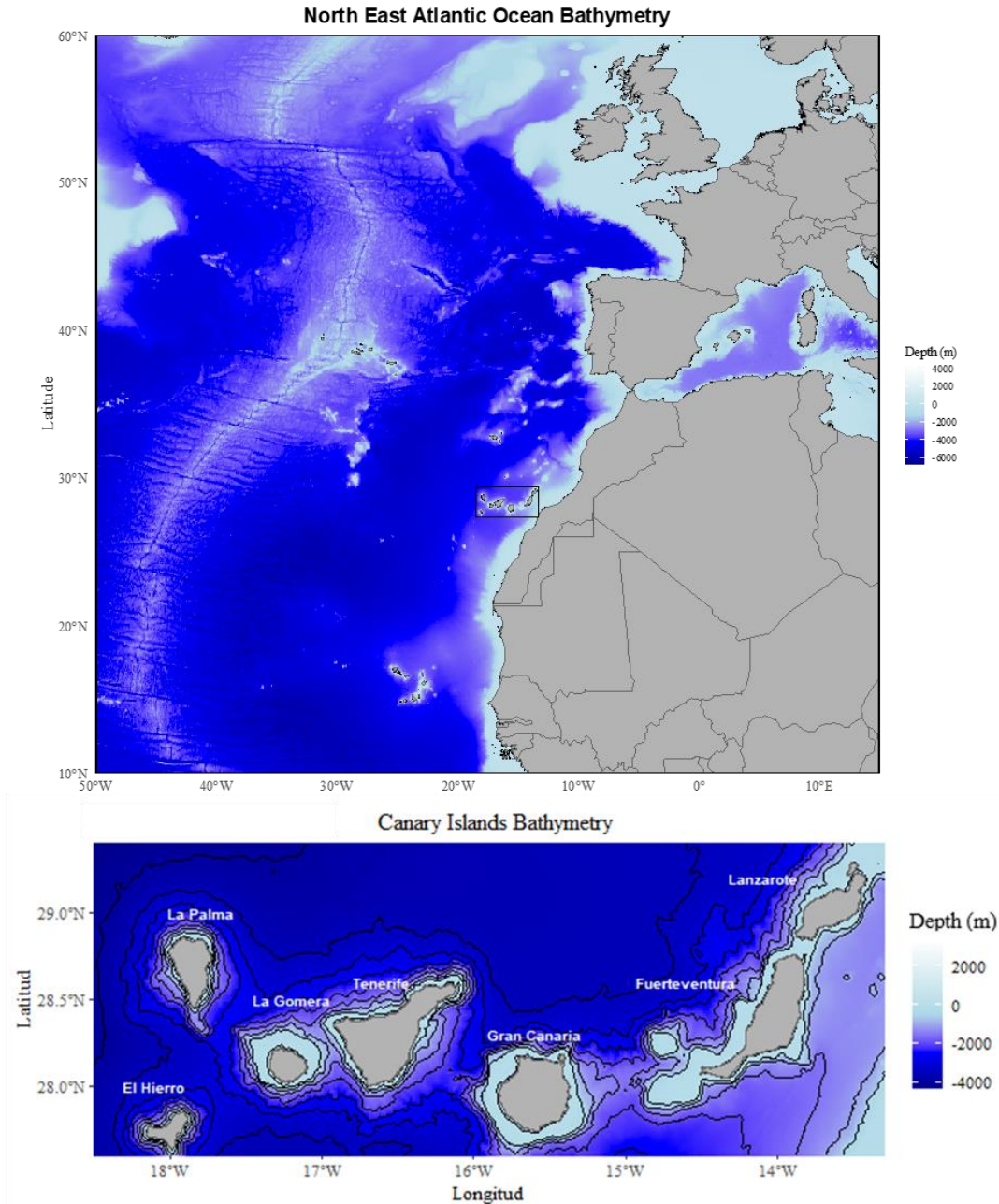


Figure 1. Bathymetric map of the Northeast Atlantic Ocean and the Canary Islands: top panel displays the bathymetry of the Northeast Atlantic Ocean, with contour lines highlighting major underwater structures; Bottom panel provides a detailed view of the Canary Islands region, illustrating bathymetric gradients and the main islands of the archipelago. Colors represent depth, ranging from deep areas (dark blue) to emerged land (gray). Labels indicate the names of the main islands (NOAA, 2009).

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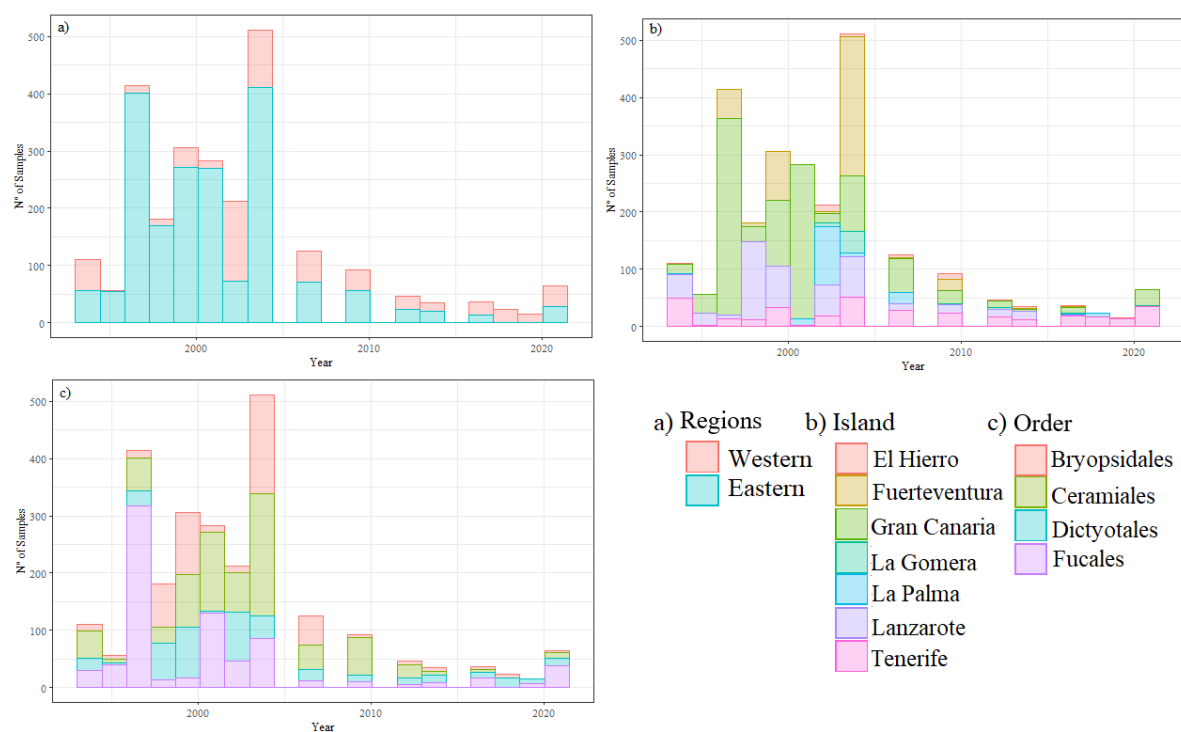


Figure 2. Histograms of algae records: (a) distribution across the Canary Islands (El Hierro, La Palma, La Gomera, Tenerife, Gran Canaria, Lanzarote, and Fuerteventura); (b) grouped by island region (Western and Eastern); and (c) by selected orders (Bryopsidales, Ceramiales, Dictyotales, and Fucales).

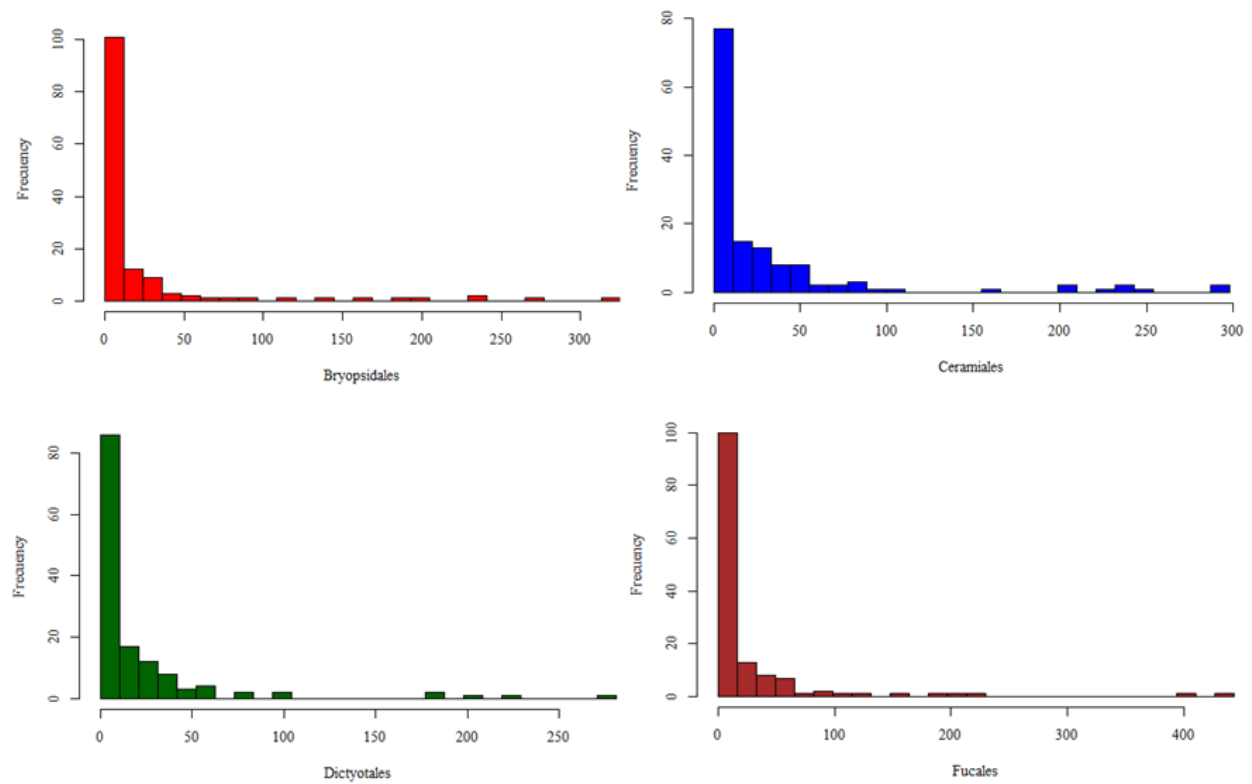


Figure. 3. Frequency diagrams of annual records for the four studied macroalgae orders over a 28-year period (1992–2020): Bryopsidales (Red), Ceramiales (Blue), Dictyotales (Green), and Fucales (Brown).

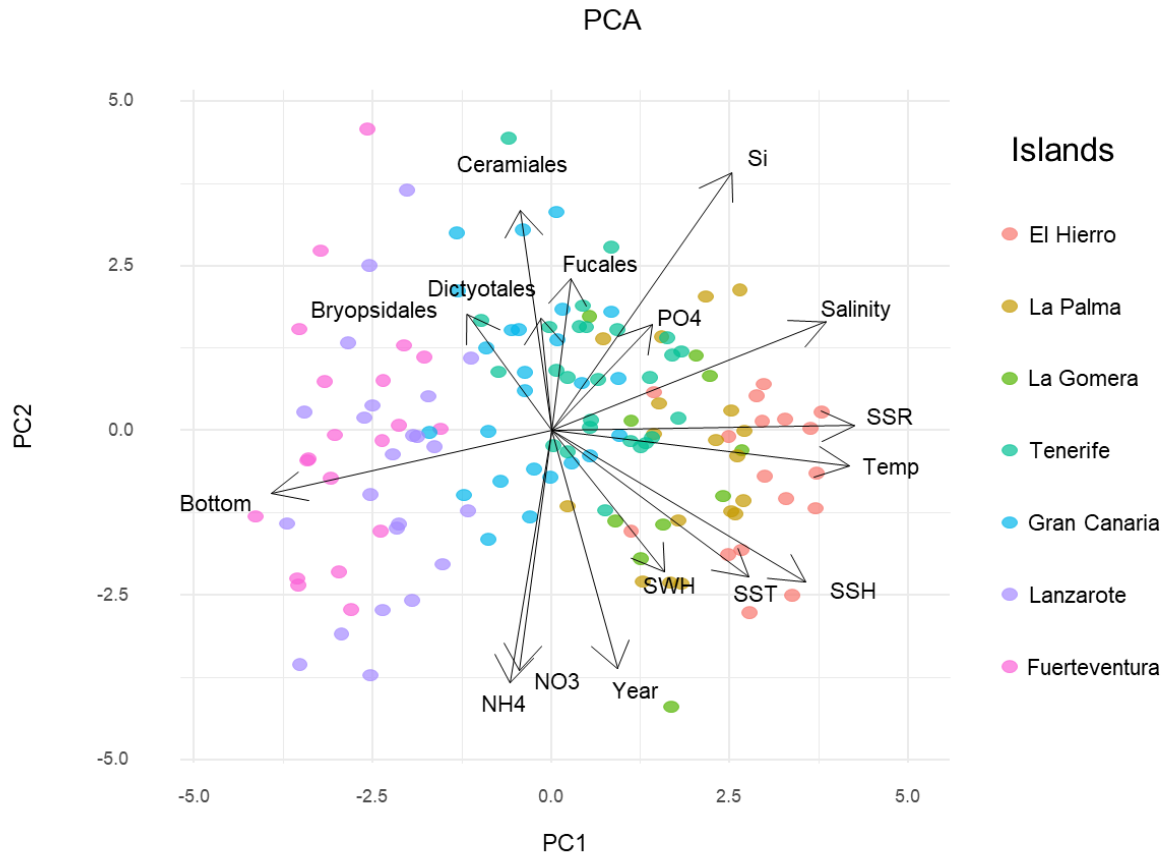


Figure 4. Principal Component Analysis (PCA) of presence-absence records for algae species (order Bryopsidales, Ceramiales, Dictyotales, and Fucales) and environmental time series data collected over 18 years (1992–2020). Environmental variables include nutrient surface concentrations ($\mu\text{g}/\text{km}^2$) for NH_4 , NO_3 , and PO_4 ; as well as environmental parameters such as temperature ($^{\circ}\text{C}$), seawater salinity, sea surface radiation (SSR), sea surface height (SSH), depth, significant wave height (SWH), and sampling year.

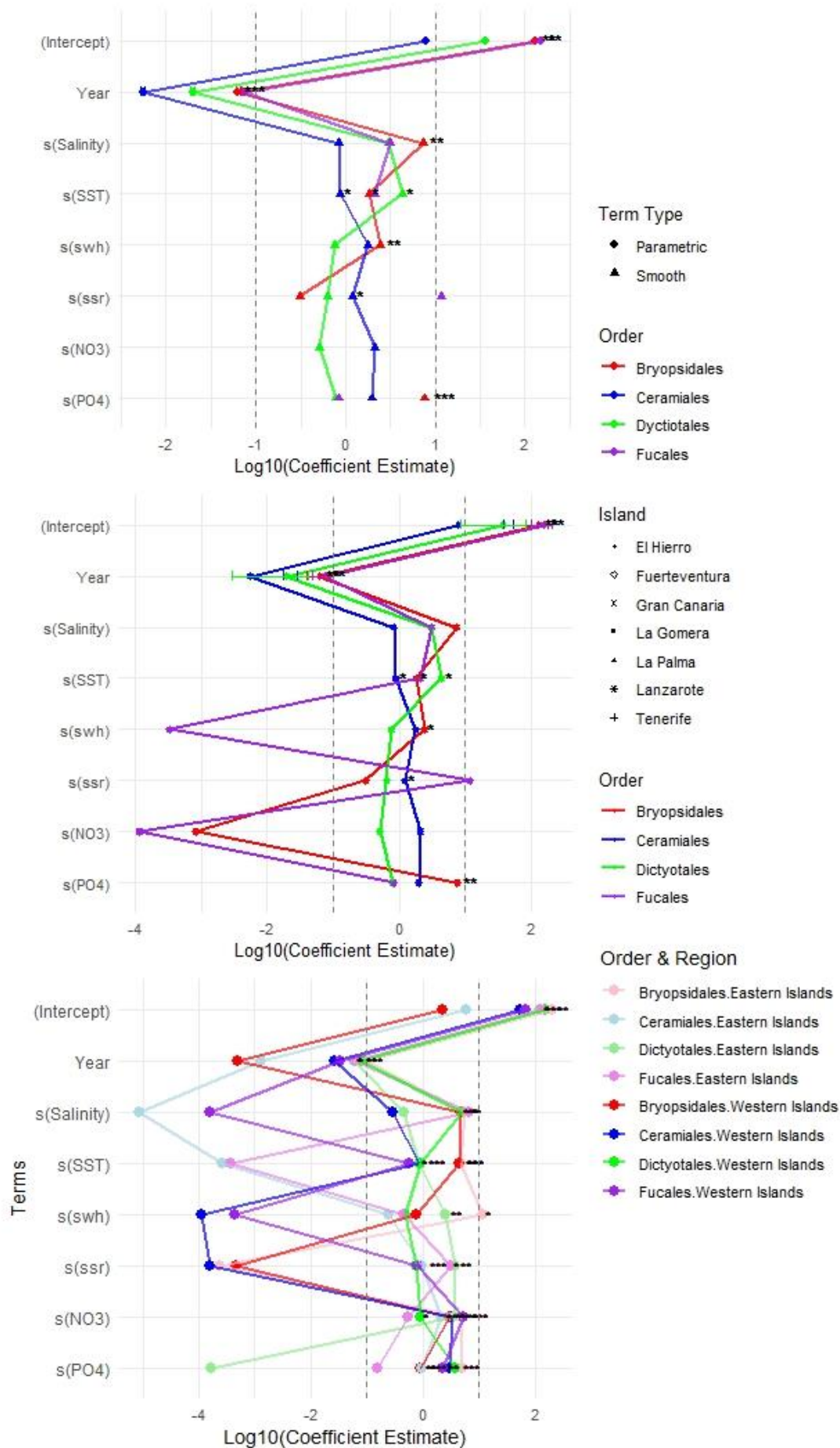


Figure 5. Log-transformed coefficient estimates from GAM models fitted to predictor variables influencing the distribution of different orders of marine algae in different perspectives of the Canary Islands, comparing the effects of environmental variables across different algal orders and regions. **Top Panel:** Effects of environmental variables on algal orders (Bryopsidales, Ceramiales, Dictyotales, and Fucales) across parametric and smooth terms. Term types are represented by different shapes (diamonds for parametric, triangles for smooth). **Middle Panel:** Island-level differences in environmental variable effects on algal orders. Each point represents the estimated coefficient for a given environmental variable across different islands. **Bottom Panel:** Regional differences in the effects of environmental variables on algal orders. Lines and points distinguish between estimates for Western Islands and Eastern Islands, with orders differentiated by color and region-order combinations indicated in the legend. Dashed vertical lines at -1 and 1 indicate thresholds for interpretation of the Log10(Coefficient Estimate). Stars indicate the significance level of terms (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$).